Cloud-containerization-and-DevOps

Experiment 10: Working with SonarQube

Introduction

What is SonarQube?

SonarQube is and open-source platform for continous inspection of code quality. It perform automatic reviews with static analysis to detect bugs, code smells &/ security vulnerabilities.

How SonarQube Solves the Problem:

- **Continous Inspection**: Scans code with everey commit, providing immediate feedback.
- **Quality Gates:** Defines pass/fail criteria for code quality.
- **Technical Debt Quantification:** Measures effort needed to fiz issues.
- **Multi-language Support:** Supports 20+ programming languages.
- **Visual Analytics:** Dashboard showing code quality metrics and trends.

Hands-On

Step-1:- Create a `docker-compose.yml`

```
nano docker-compose.yml
```

Paste This:

```yaml
version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
```

```yaml
    volumes:
      - sonar-db-data:/var/lib/postgresql/data
    networks:
      - sonarqube-lab

  sonarqube:
    image: sonarqube:lts-community
    container_name: sonarqube
    ports:
      - "9000:9000"
    environment:
      SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube
      SONAR_JDBC_USERNAME: sonar
      SONAR_JDBC_PASSWORD: sonar
    volumes:
      - sonar-data:/opt/sonarqube/data
      - sonar-extensions:/opt/sonarqube/extensions
    depends_on:
      - sonar-db
    networks:
      - sonarqube-lab

volumes:
  sonar-db-data:
  sonar-data:
  sonar-extensions:

networks:
  sonarqube-lab:
    driver: bridge
```

```
palakagarwal_16@LAPTOP-2S
  GNU nano 7.2                    docker-compose.yml
version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    restart: unless-stopped
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
    volumes:
      - sonar-db-data:/var/lib/postgresql/data
    networks:
      - sonarqube-lab

  sonarqube:
    image: sonarqube:lts-community
    container_name: sonarqube
    restart: unless-stopped
    ports:
      - "9000:9000"
    environment:
      SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube
      SONAR_JDBC_USERNAME: sonar
      SONAR_JDBC_PASSWORD: sonar
    volumes:
      - sonar-data:/opt/sonarqube/data
      - sonar-extensions:/opt/sonarqube/extensions
    depends_on:
      - sonar-db
    networks:
      - sonarqube-lab

volumes:
  sonar-db-data:
  sonar-data:
```

Step-2:- Start Compose

```
docker-compose up -d
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/sample-java-app$ docker comp
ose up -d
WARN[0000] /mnt/c/Users/hp/docker-compose.yml: the attribute `version` is ob
solete, it will be ignored, please remove it to avoid potential confusion
WARN[0000] Found orphan containers ([ba15d4b7600d]) for this project. If you
 removed or renamed this service in your compose file, you can run this comm
and with the --remove-orphans flag to clean it up.
[+] up 2/2
 ✔Container sonar-db  Running                                          0.0s
 ✔Container sonarqube Running                                          0.0s
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/sample-java-app$
```

Step-3:- Verify SonarQube is running via Logs

```
docker-compose logs -f sonarqube
```

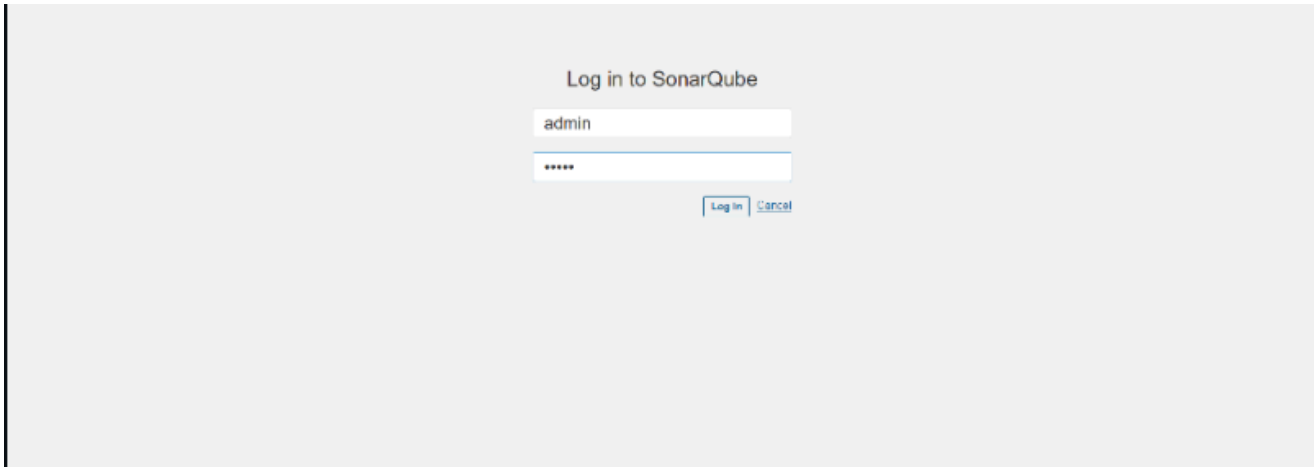
```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ docker logs -f sonarqube
2026.04.18 04:22:35 INFO  app[][o.s.a.AppFileSystem] Cleaning or creating te
mp directory /opt/sonarqube/temp
2026.04.18 04:22:35 INFO  app[][o.s.a.es.EsSettings] Elasticsearch listening
 on [HTTP: 127.0.0.1:9001, TCP: 127.0.0.1:35219]
2026.04.18 04:22:35 INFO  app[][o.s.a.ProcessLauncherImpl] Launch process[EL
ASTICSEARCH] from [/opt/sonarqube/elasticsearch]: /opt/sonarqube/elasticsear
ch/bin/elasticsearch
2026.04.18 04:22:35 INFO  app[][o.s.a.SchedulerImpl] Waiting for Elasticsear
ch to be up and running
2026.04.18 04:22:44 INFO  es[][o.e.n.Node] version[7.17.15], pid[25], build[
default/tar/0b8ecfb4378335f4689c4223d1f1115f16bef3ba/2023-11-10T22:03:46.987
399016Z], OS[Linux/6.6.87.2-microsoft-standard-WSL2/amd64], JVM[Eclipse Adop
tium/OpenJDK 64-Bit Server VM/17.0.16/17.0.16+8]
2026.04.18 04:22:44 INFO  es[][o.e.n.Node] JVM home [/opt/java/openjdk]
2026.04.18 04:22:44 INFO  es[][o.e.n.Node] JVM arguments [-XX:+UseG1GC, -Dja
va.io.tmpdir=/opt/sonarqube/temp, -XX:ErrorFile=/opt/sonarqube/logs/es_hs_er
r_pid%p.log, -Des.networkaddress.cache.ttl=60, -Des.networkaddress.cache.neg
ative.ttl=10, -XX:+AlwaysPreTouch, -Xss1m, -Djava.awt.headless=true, -Dfile.
encoding=UTF-8, -Djna.nosys=true, -Djna.tmpdir=/opt/sonarqube/temp, -XX:-Omi
tStackTraceInFastThrow, -Dio.netty.noUnsafe=true, -Dio.netty.noKeySetOptimiz
ation=true, -Dio.netty.recycler.maxCapacityPerThread=0, -Dio.netty.allocator
.numDirectArenas=0, -Dlog4j.shutdownHookEnabled=false, -Dlog4j2.disable.jmx=
true, -Dlog4j2.formatMsgNoLookups=true, -Djava.locale.providers=COMPAT, -Dco
m.redhat.fips=false, -Des.enforce.bootstrap.checks=true, -Xmx512m, -Xms512m,
 -XX:MaxDirectMemorySize=256m, -XX:+HeapDumpOnOutOfMemoryError, -Des.path.ho
me=/opt/sonarqube/elasticsearch, -Des.path.conf=/opt/sonarqube/temp/conf/es,
 -Des.distribution.flavor=default, -Des.distribution.type=tar, -Des.bundled_
jdk=false]
2026.04.18 04:22:45 INFO  es[][o.e.p.PluginsService] loaded module [analysis
-common]
2026.04.18 04:22:45 INFO  es[][o.e.p.PluginsService] loaded module [lang-pai
nless]
2026.04.18 04:22:45 INFO  es[][o.e.p.PluginsService] loaded module [parent-j
oin]
2026.04.18 04:22:45 INFO  es[][o.e.p.PluginsService] loaded module [reindex]
2026.04.18 04:22:45 INFO  es[][o.e.p.PluginsService] loaded module [transpor
t-netty4]
2026.04.18 04:22:45 INFO  es[][o.e.p.PluginsService] no plugins loaded
```

Step-4:- Login SonarQube

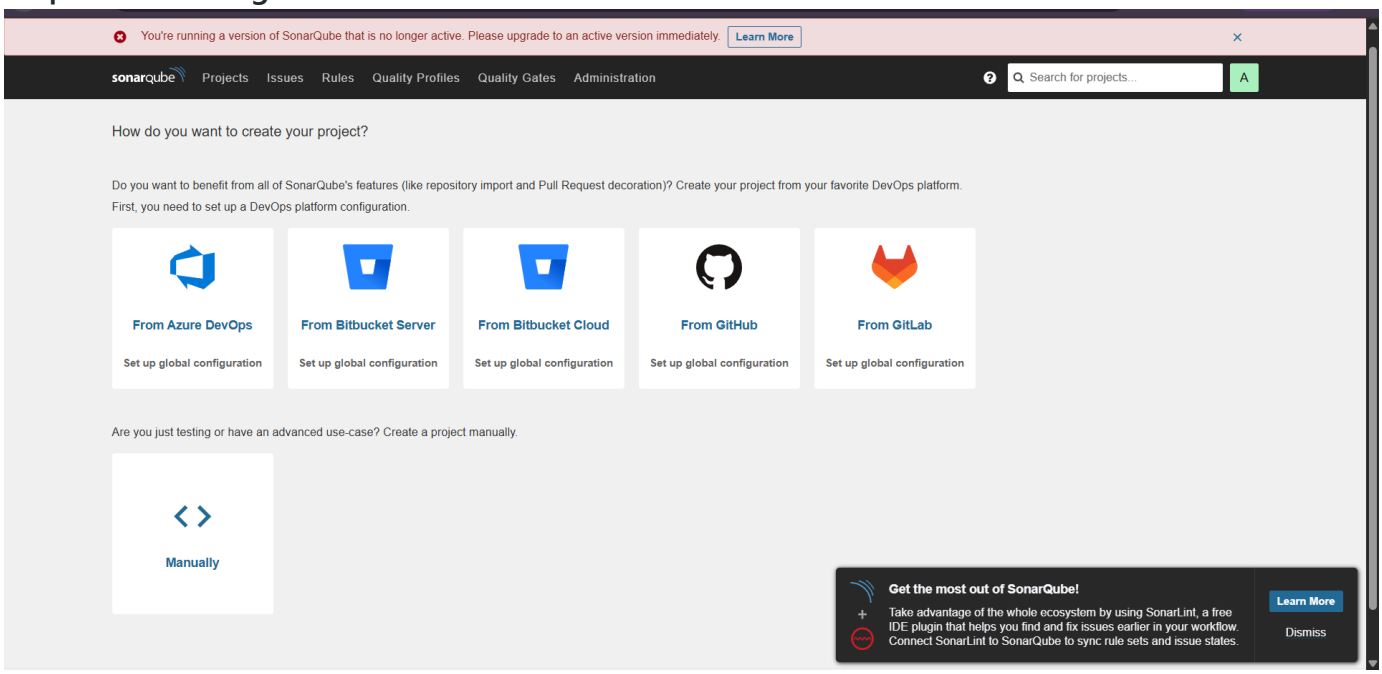
```
http://localhost:9001
```

- username: admin

- password: admin



Step-5:- HomePage



Step-6:- Create Sample Java App

```
mkdir -p sample-java-app/src/main/java/com/example
cd sample-java-app
```

Step-7:- Create `Calculator.java`

```
nano src/main/java/com/example/Calculator.java
```

```
package com.example;

public class Calculator {

    public int divide(int a, int b) {
        return a / b;
    }
```

```java
    public int add(int a, int b) {
        int result = a + b;
        int unused = 100;
        return result;
    }

    public String getUser(String userId) {
        String query = "SELECT * FROM users WHERE id = " + userId;
        return query;
    }

    public int multiply(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public int multiplyAlt(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public String getName(String name) {
        return name.toUpperCase();
    }

    public void riskyOperation() {
        try {
            int x = 10 / 0;
        } catch (Exception e) {
        }
    }
}
```

```java
package com.example;

public class Calculator {

    public int divide(int a, int b) {
        return a / b;
    }

    public int add(int a, int b) {
        int result = a + b;
        int unused = 100;
        return result;
    }

    public String getUser(String userId) {
        String query = "SELECT * FROM users WHERE id = " + userId;
        return query;
    }

    public int multiply(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public int multiplyAlt(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public String getName(String name) {
        return name.toUpperCase();
```

Step-8:- Create `pom.xml`

```
nano pom.xml
```

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>sample-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <sonar.projectKey>sample-java-app</sonar.projectKey>
        <sonar.host.url>http://localhost:9000</sonar.host.url>
        <sonar.login>YOUR_TOKEN</sonar.login>
    </properties>
```

```
    <build>
        <plugins>
            <plugin>
                <groupId>org.sonarsource.scanner.maven</groupId>
                <artifactId>sonar-maven-plugin</artifactId>
                <version>3.9.1.2184</version>
            </plugin>
        </plugins>
    </build>
</project>
```

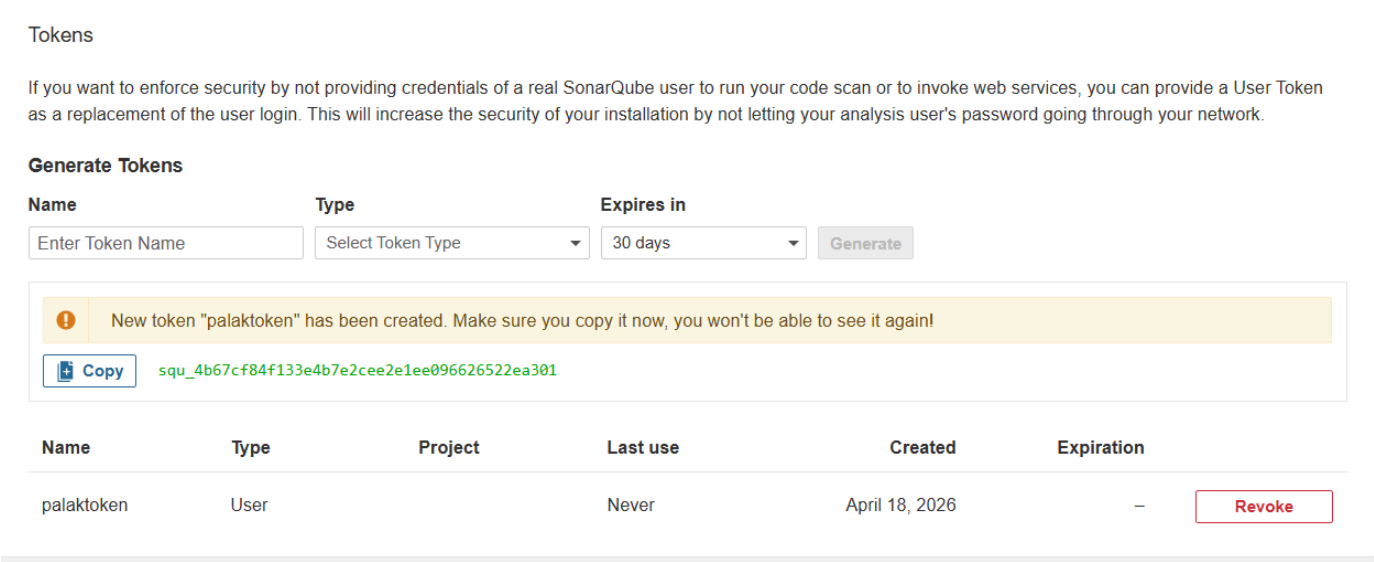
```
<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>sample-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <sonar.projectKey>sample-java-app</sonar.projectKey>
        <sonar.host.url>http://localhost:9000</sonar.host.url>
        <sonar.login>YOUR_TOKEN</sonar.login>
    </properties>

    <build>
        <plugins>
            <plugin>
                <groupId>org.sonarsource.scanner.maven</groupId>
                <artifactId>sonar-maven-plugin</artifactId>
                <version>3.9.1.2184</version>
            </plugin>
        </plugins>
    </build>
</project>
```

Step-9:- Create Tokens Profile -> My Account -> Security -> TokenName: `scanner-token` -> Generate



Step-10:- Run Scanner

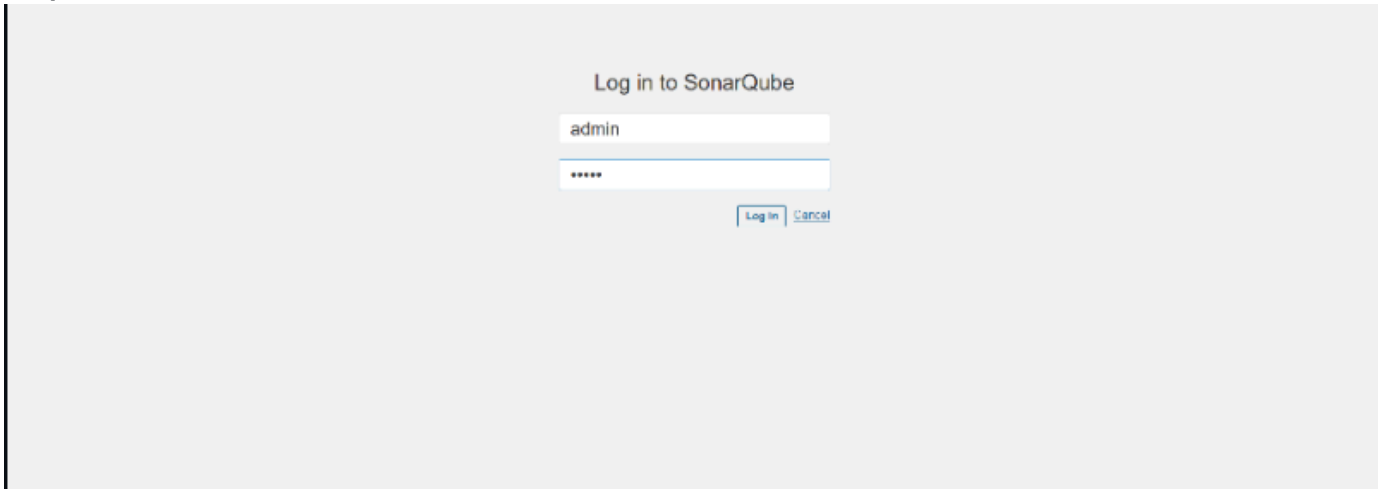
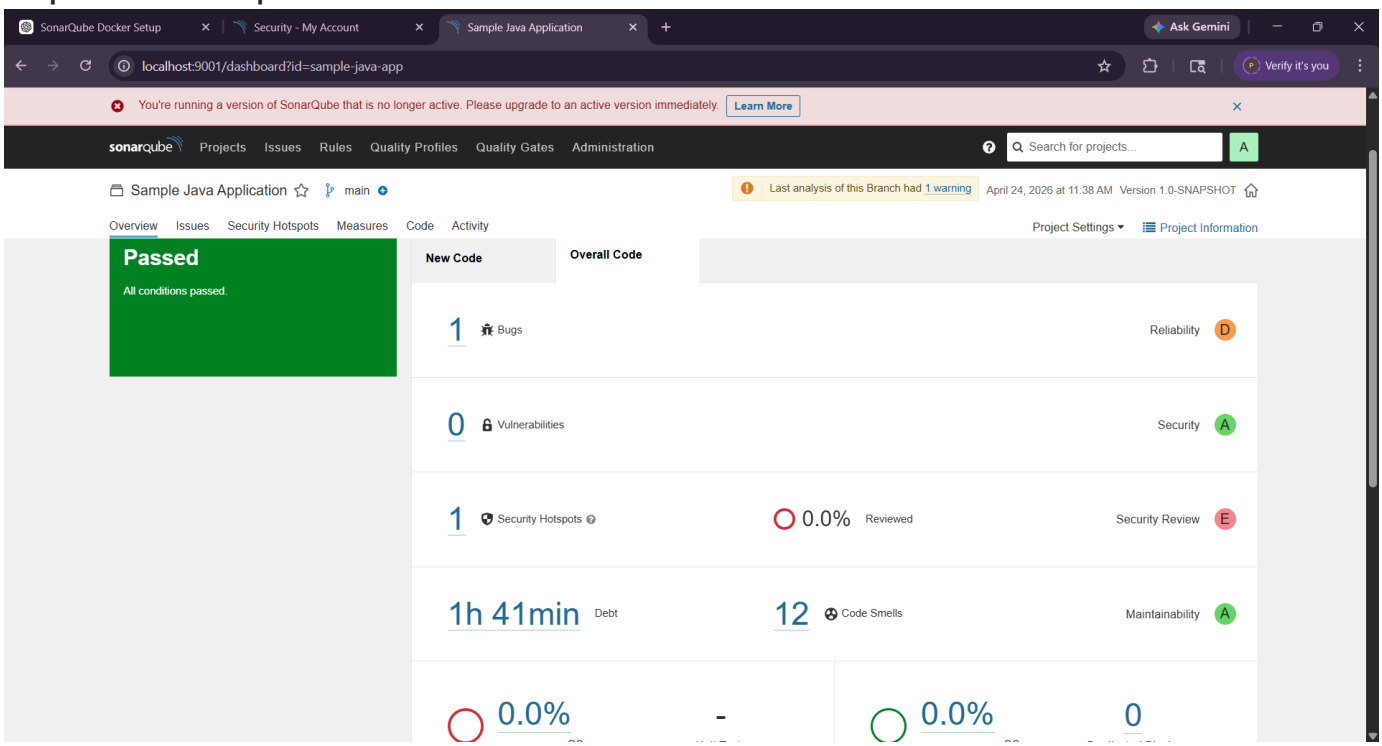
```
mvn sonar:sonar -Dsonar.login=YOUR_TOKEN
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/sample-java-app$ mvn sonar:s
onar -Dsonar.login=squ_2e95439e05120daf211c16a731ece5545447a39f
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:sample-app >--------------------
---
[INFO] Building sample-app 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
---
[INFO]
[INFO] --- sonar-maven-plugin:4.0.0.4121:sonar (default-cli) @ sample-app --
-
[INFO] 06:08:09.532 Java 21.0.10 Ubuntu (64-bit)
[INFO] 06:08:09.533 Linux 6.6.87.2-microsoft-standard-WSL2 (amd64)
[INFO] 06:08:10.646 User cache: /home/palakagarwal_16/.sonar/cache
[INFO] 06:08:11.249 Communicating with SonarQube Server 9.9.8.100196
[INFO] 06:08:11.394 Default locale: "en", source code encoding: "UTF-8" (ana
lysis is platform dependent)
[INFO] 06:08:13.167 Load global settings
[INFO] 06:08:13.678 Load global settings (done) | time=515ms
[INFO] 06:08:13.689 Server id: 74C15348-AZ2-FeoTJ4JgJc_ZF7TJ
[INFO] 06:08:13.708 User cache: /home/palakagarwal_16/.sonar/cache
[INFO] 06:08:13.728 Load/download plugins
[INFO] 06:08:13.728 Load plugins index
[INFO] 06:08:13.928 Load plugins index (done) | time=200ms
[INFO] 06:08:20.989 Load/download plugins (done) | time=7261ms
[INFO] 06:08:22.617 Process project properties
[INFO] 06:08:22.644 Process project properties (done) | time=27ms
[INFO] 06:08:22.653 Execute project builders
[INFO] 06:08:22.656 Execute project builders (done) | time=3ms
[INFO] 06:08:22.671 Project key: sample-java-app
[INFO] 06:08:22.672 Base dir: /mnt/c/Users/hp/sample-java-app
[INFO] 06:08:22.672 Working dir: /mnt/c/Users/hp/sample-java-app/target/sona
r
[INFO] 06:08:22.704 Load project settings for component key: 'sample-java-ap
p'
[INFO] 06:08:23.525 Load quality profiles
[INFO] 06:08:23.853 Load quality profiles (done) | time=329ms
[INFO] 06:08:23.891 Load active rules
[INFO] 06:08:29.076 Load active rules (done) | time=5185ms
```

Step-11:- Scroll down to view Build Success

```
[INFO] 06:08:42.267 Sensor Analysis Warnings import [csharp] (done) | time=2
ms
[INFO] 06:08:42.268 Sensor Zero Coverage Sensor
[INFO] 06:08:42.303 Sensor Zero Coverage Sensor (done) | time=35ms
[INFO] 06:08:42.304 Sensor Java CPD Block Indexer
[INFO] 06:08:42.355 Sensor Java CPD Block Indexer (done) | time=51ms
[INFO] 06:08:42.365 SCM Publisher SCM provider for this project is: git
[INFO] 06:08:42.368 SCM Publisher 2 source files to be analyzed
[WARNING] 06:08:42.613 Thread[#50,JGit-FileStoreAttributeReader-1,5,main]: g
t smaller file timestamp on /mnt/c (C:\), /mnt/c/Users/hp/.git: 2026-04-24T
06:08:42Z < 2026-04-24T06:08:42.1555933Z. Aborting measurement at resolution
PT0.8444067S.
[INFO] 06:08:42.867 SCM Publisher 0/2 source files have been analyzed (done)
| time=497ms
[WARNING] 06:08:42.867 Missing blame information for the following files:
[WARNING] 06:08:42.868   * src/main/java/com/example/Calculator.java
[WARNING] 06:08:42.868   * pom.xml
[WARNING] 06:08:42.869 This may lead to missing/broken features in SonarQube
[INFO] 06:08:42.876 CPD Executor Calculating CPD for 1 file
[INFO] 06:08:42.897 CPD Executor CPD calculation finished (done) | time=20ms
[INFO] 06:08:43.570 Analysis report generated in 365ms, dir size=130.7 kB
[INFO] 06:08:45.208 Analysis report compressed in 1624ms, zip size=20.4 kB
[INFO] 06:08:46.220 Analysis report uploaded in 1011ms
[INFO] 06:08:46.232 ANALYSIS SUCCESSFUL, you can find the results at: http:/
/localhost:9001/dashboard?id=sample-java-app
[INFO] 06:08:46.232 Note that you will be able to access the updated dashboa
rd once the server has processed the submitted analysis report
[INFO] 06:08:46.233 More about the report processing at http://localhost:900
1/api/ce/task?id=AZ2-GuJeJ4JgJc_ZGFMj
[INFO] 06:08:46.494 Analysis total time: 25.115 s
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  37.802 s
[INFO] Finished at: 2026-04-24T06:08:46Z
[INFO] ------------------------------------------------------------------------
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/sample-java-app$ docker ps
```

Step-12:- On Browser test will be Listed



Step-13:- View Report